

GEORGIA STATE UNIVERSITY Robinson College of Business IFI 8410: Programming for Business Syllabus: Spring 2026

**This Course Syllabus Provides a General Plan for the Course;
Deviations May Be Necessary**

INSTRUCTOR:

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COURSE DESCRIPTION

This course introduces programming as a practical approach to business data analysis and decision support. Students learn how programming facilitates the development of flexible, scalable, and reproducible analytical workflows for addressing data-driven business questions in organizational contexts.

The course emphasizes the role of programming in structuring and executing end-to-end analytics workflows, including data preparation, exploratory analysis, basic modeling, and interpretation of results. Python is used as the primary programming environment to implement these concepts.

Through a combination of lectures, in-class programming exercises, assignments, and a team-based term project, students apply programming to real-world business problems and develop the ability to communicate analytical insights effectively to managerial audiences.

PREREQUISITES

There are no formal prerequisites for this course.

LEARNING OBJECTIVES

Upon successful completion of this course, students will be able to:

1. Explain the role of programming in business analytics, including how programming supports flexible, scalable, and reproducible analytical workflows.
2. Apply core programming concepts using Python to prepare, explore, and analyze business data.
3. Design and execute end-to-end analytics workflows that address data-driven business questions in organizational contexts.
4. Select and apply appropriate analytical techniques based on the characteristics of the data and the business problem.
5. Interpret and communicate analytical results effectively to managerial audiences, including discussing assumptions, limitations, and implications for decision-making.

6. Collaborate effectively in team-based analytical projects, demonstrating professional responsibility, contribution, and peer accountability.

COURSE FORMAT

This course is delivered in a combined lecture and applied learning format. Class sessions integrate conceptual discussion with hands-on programming demonstrations and guided practice. Programming concepts and analytical techniques are introduced in class and reinforced through in-class exercises, assignments, and a team-based term project.

Students are expected to actively engage during class sessions by completing in-class programming activities, participating in structured exercises, and contributing to project-related discussions. These activities are designed to support continuous learning and practical skill development.

Outside of class, students are expected to review course materials, practice programming skills, complete assignments, and work collaboratively on the term project. The course is cumulative in nature; consistent engagement throughout the semester is important for success.

The instructor will provide guidance, feedback, and clarification throughout the semester, including structured opportunities for progress check-ins related to the term project.

COURSE MATERIALS

There is **no required textbook** for this course. All required readings, examples, and instructional materials will be provided through *iCollege*.

To support learning and provide additional context, several recommended resources are listed below. These materials are intended as references and supplements; students are **not required** to purchase or read them in full. They are particularly useful for students who wish to deepen their understanding of programming, data analysis, and analytical thinking in business contexts.

Recommended Books and Online Resources

- McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter* (3rd ed.). O'Reilly Media. <https://wesmckinney.com/book/>
- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in Python*. Springer. <https://www.statlearning.com/>
- W3Schools. *Python tutorial*. <https://www.w3schools.com/python/default.asp>

Additional materials may be assigned during the semester to illustrate specific concepts or applications. Any such materials will be announced in advance.

SOFTWARE AND TOOLS

This course uses **Python** as the primary programming language and **Google Colab** as the main computing environment.

Google Colab is a browser-based platform that enables students to write, execute, and document Python code without installing software locally. It supports interactive notebooks that integrate code, output, and narrative text, and provides access to commonly used Python libraries for data

analysis. All course materials, examples, and in-class exercises will be demonstrated using Google Colab.

Students who prefer to use an alternative development environment (e.g., VS Code, Jupyter Notebook, or other local Python setups) may do so, provided that their work is fully compatible with the course requirements and can be submitted in the specified formats. Students using alternative environments are responsible for managing their own setup and ensuring that their code runs correctly.

As class sessions include hands-on programming exercises, students are expected to have access to a laptop or equivalent computing device during class to actively participate.

GRADING AND EVALUATION

Student performance is evaluated based on assignments, in-class engagement, and a team-based term project. The assessment structure is designed to emphasize **applied programming practice**, analytical reasoning, and effective communication of insights in business contexts.

The relative weights of the course components are as follows:

Component	Weight
Assignments ($3 \times 10\%$)	30%
Attendance	10%
Participation	10%
Term Project	50%
Project Presentation	40%
Peer Evaluation	10%
Total	100%

Assignments (30%)

Assignments consist primarily of individual coding exercises and short programming tasks designed to reinforce concepts introduced in class and to build skills required for the term project.

Term Project (50%)

The term project is the central integrative component of the course. Projects are typically completed in teams of **up to three students**. Students may choose to complete the project **individually**, subject to **prior approval from the instructor**.

Teams (or approved individuals) design and execute an end-to-end analytics workflow addressing a business-relevant question using real data.

The project is evaluated primarily through **a final presentation (40%)**, which assesses:

- Clarity of the business problem and motivation
- Appropriateness of data and analytical approach
- Interpretation of results

- Managerial implications and limitations

There is **no written project report**. Teams must submit a complete and reproducible Jupyter notebook as supporting material; however, the notebook is required but is **not graded separately**.

Peer evaluation (10%) is used to assess individual contribution within teams and may be used to adjust individual project scores.

Attendance Policy (10%)

Attendance reflects students' presence in scheduled class sessions. Attendance will be recorded each session. Absences due to documented, university-approved circumstances may be accommodated at the instructor's discretion.

Participation (10%)

Participation reflects active engagement during class sessions and is evaluated based on **effort and engagement**, not on correctness or frequency of speaking. Participation may include, but is not limited to:

- Completion of in-class programming exercises
- Participation in low-stakes quizzes, polls, or concept checks
- Project-related check-ins or brief reflections
- Engagement in structured class activities

Participation activities conducted during scheduled class sessions cannot be made up.

Grading Scale

Final letter grades will be assigned according to the following scale:

Letter Grade	Percentage
A+	97.0 – 100
A	93.0 – 96.9
A–	90.0 – 92.9
B+	87.0 – 89.9
B	83.0 – 86.9
B–	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
C–	70.0 – 72.9
D	60.0 – 69.9
F	0 – 59.9

A grade of 60 or above is considered passing. An **A+** is reserved for exceptional performance that demonstrates outstanding integration of course concepts, analytical rigor, and clarity of interpretation beyond standard course expectations.

Course assignments and project deliverables, along with their due dates, are specified in the Tentative Schedule and in iCollege. Students are responsible for tracking deadlines and submitting all work on time.

DEADLINES AND LATE SUBMISSION POLICY

All deadlines are **strict**. Unless otherwise noted, all assignments and project deliverables are due at **23:59 (ET)** on the stated date.

Late submissions are **not accepted** except in cases of documented, university-approved circumstances. Technical issues, competing commitments, or misinterpretation of deadlines do not constitute valid reasons for late submission.

MAKE-UP WORK POLICY

Make-up assignments, alternative participation opportunities, or substitute project deliverables are not provided, except in cases of documented, university-approved circumstances. Students should notify the instructor as early as possible if such circumstances arise.

GENERAL UNIVERSITY POLICIES

Course Assessment

Your constructive assessment of this course plays an indispensable role in shaping education at Georgia State. Upon completing the course, please take the time to fill out the online course evaluation.

Ethics and Academic Honesty

Students are expected to adhere to the highest standards of academic integrity in all course activities. All submitted work must reflect each student's own understanding and effort, except where collaboration is explicitly permitted.

Academic honesty violations include, but are not limited to:

- Plagiarism (see course and GSU website for specific examples of what constitutes plagiarism),
- Unauthorized collaboration
- Submitting work generated by AI tools without sufficient understanding or attribution
- Representing others' work as one's own

Violations of academic honesty will be handled in accordance with Georgia State University policies.

Use of AI Tools

AI-based tools (e.g., ChatGPT, Copilot) may be used **as learning aids**, such as clarifying concepts, exploring ideas, or debugging code. However, AI tools may not replace a student's own reasoning,

analysis, or understanding in any graded work.

Students are responsible for ensuring that all submitted work reflects their own comprehension and effort and must be able to explain and justify all results, code, and conclusions.

Special Needs

Students who wish to request accommodation for a disability may do so by registering with the Access and Accommodations Center. Students may only be accommodated upon issuance by the Access and Accommodation Center of a signed Accommodation Plan and are responsible for providing a copy of that plan to instructors of all classes in which accommodations are sought.

Family Educational Rights and Privacy Act

The Family Educational Rights and Privacy Act (FERPA) protects the privacy of student education records. In accordance with FERPA, the instructor cannot discuss a student's academic performance or course records with anyone other than the student, unless written consent has been provided or disclosure is permitted by law.

Sexual Harassment and Title IX

Georgia State University is committed to maintaining a learning environment free from discrimination and harassment. Sexual harassment, including sexual violence, is prohibited under Title IX.

Students who experience or witness sexual harassment or misconduct are encouraged to seek support and report concerns through appropriate university resources. Information regarding reporting options and support services is available through Georgia State University's Equal Opportunity and Title IX resources.

Basic Needs Support

Georgia State University recognizes that students may experience challenges related to food, housing, financial security, or other basic needs that may affect academic performance and well-being. Students facing such challenges are encouraged to reach out to campus resources or to contact the instructor to discuss available support options.

Unauthorized Public Posting and Distribution of Course Materials

All materials created for this course, including but not limited to lecture slides, notebooks, assignments, handouts, and recorded content, are intended solely for use by students enrolled in this course for educational purposes.

Course materials may not be shared with individuals outside the class or posted on public or external platforms without explicit permission from the instructor. Unauthorized distribution constitutes a violation of course rules and may be treated as a violation of university policies.

TENTATIVE SCHEDULE

Adjustments may be made; Additional materials may be assigned.

MODULE 0: Course Introduction and Orientation	
01/14	Course Overview and Programming for Business • Course orientation, structure, and expectations • Student introductions and course community building • Programming for business problem solving • Overview of programming languages and the role of Python • Introduction to the programming environment (<i>Google Colab</i>) • In-class demonstration and guided practice
MODULE 1: Programming Foundations for Business Applications	
01/21	Python Programming Basics and Interactive Computing • Python execution model and interactive computing • Variables, basic data types, and expressions • Programming for business problem solving • Overview of essential Python libraries for data-oriented work • In-class demonstration and guided practice Deliverable: Teams formed by 01/21, 23:59
01/28	Control Flow, Functions, and Core Data Structures • Conditional logic and iteration for business problem solving • Writing and using functions to structure programs • Core built-in data structures (lists, tuples, dictionaries) • Representing simple business data programmatically • In-class demonstration and guided practice
02/04	NumPy Foundations and Working with External Data • NumPy arrays and vectorized computation • Relationship between NumPy and higher-level data tools • Reading data from external files (e.g., CSV, Excel, TXT) • Inspecting and understanding imported data • In-class demonstration and guided practice Note: Assignment 1 released after class Deliverable: Assignment 1 due 02/10, 23:59

MODULE 2: Data Preparation and Exploration	
02/11	Introduction to pandas and Data Cleaning • pandas data structures and core functionality • Identifying and addressing common data quality issues • Transforming variables programmatically • In-class demonstration and guided practice
02/18	Data Wrangling, Combining, and Aggregation • Combining datasets using joins and merges • Concatenating and reshaping data • Grouping and aggregation for summarization • Working with panel-style data structures • In-class demonstration and guided practice
02/25	Descriptive Analysis and Visualization • Descriptive statistics for understanding data • Exploring distributions, relationships, and patterns • Visualization for data understanding and communication • Identifying anomalies and potential data quality concerns • In-class demonstration and guided practice Deliverables: Project Step 1 due 02/24, 23:59
03/04	Working with Time-Oriented and Text Data • Working with date and time information in data • Handling time-oriented and longitudinal data • Introductory processing of text data • In-class demonstration and guided practice Note: Assignment 2 released after class Deliverables: Assignment 2 due 03/10, 23:59
03/11	Project work session • No formal class meeting
03/18	Spring Break: 03/16 – 03/22 (No Class)

MODULE 3: Modeling and Applied Programming	
03/25	Introduction to Machine Learning with Python • What machine learning is and is not in business contexts • Prediction vs. explanation for managerial decision-making • Supervised vs. unsupervised learning (conceptual overview) • Intuition behind model training and evaluation • In-class demonstration and guided practice
04/01	Supervised Learning I: Regression • Business problems involving continuous outcomes • Regression as a modeling approach for prediction and explanation • Interpreting regression outputs and assumptions at a high level • Understanding model performance intuitively • In-class demonstration and guided practice
04/08	Supervised Learning II: Classification • Business problems involving categorical outcomes • Classification concepts • Interpreting classification results and performance metrics • Tradeoffs between accuracy and interpretability • In-class demonstration and guided practice Note: Assignment 3 released after class Deliverables: Assignment 3 due 04/14, 23:59
04/15	Unsupervised Learning: Clustering and Pattern Discovery • When labels are not available in business data • Clustering for segmentation and exploratory insight • Interpreting patterns without ground-truth outcomes • Practical considerations and limitations • In-class demonstration and guided practice
04/22	Advanced Modeling Methods (Overview) • Why simple models may be insufficient in complex settings • High-level overview of advanced modeling approaches • Tradeoffs between performance, complexity, and interpretability • In-class demonstration and guided practice
04/29	Final Project Presentations • Team project presentations • Peer evaluation and discussion Deliverables: • Final presentation slides due 04/27, 23:59 • Peer evaluation due 04/29, 23:59 (via iCollege)